

# AirSense Pakistan

## Hardware Architecture and Technical Prediction Framework

|                                                                                                     |                                                                                             |
|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Document Type:</b> Engineering Specification & Deployment Blueprint                              | <b>Initiative:</b> COIL AI Collaborative Research                                           |
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| <b>Institution:</b> Beaconhouse International College (BIC)                                         | <b>Release Version:</b> v3.0 Enterprise Operational Edition (August 2026)                   |

### 1. Executive Overview and Multi-Campus Objectives

AirSense Pakistan is an enterprise-grade IoT sensing and predictive machine learning platform engineered during the **COIL AI** program. The framework establishes a unified environmental intelligence infrastructure spanning Beaconhouse International College (BIC) campuses in Islamabad and Karachi, with complete architectural readiness for nationwide scaling across Lahore, Rawalpindi, Faisalabad, and Peshawar.

### 2. Edge Sensing Hardware Architecture

| Subsystem                  | Component & Model                    | Interface / Protocol                      | Key Operational Role & Engineering Specification                                                                                                          |
|----------------------------|--------------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Core Compute & Telemetry   | Espressif ESP32 WROOM-32D            | Wi-Fi 802.11 b/g/n<br>BLE 4.2 / Dual Core | Dual-core 240 MHz Xtensa LX6. Core 0 executes HTTP/MQTT cloud telemetry and ring-buffer sync. Core 1 manages deterministic sensor polling and SD logging. |
| Particulate Matter Sensing | Plantower PMS7003 Laser Counter      | UART Serial (9600 baud, 3.3V)             | Laser scattering chamber with integrated constant-flow fan. Simultaneously measures PM1.0, PM2.5, and PM10 mass concentrations (0.3 to 10 um range).      |
| Meteorological Drivers     | Bosch BME280 Environmental Sensor    | I2C Bus (Address 0x76, 3.3V)              | High-accuracy ambient temperature (+-0.5 C), relative humidity (+-3%), and barometric pressure (+-1 hPa) for boundary layer stagnation indexing.          |
| Local Offline Buffer       | SPI MicroSD Module + 16GB EVO+       | SPI Bus (CS Pin GPIO5, 3.3V)              | Circular logging buffer preserving raw 1-minute and canonical 1-hour readings locally for 180+ days, preventing data loss during campus Wi-Fi outages.    |
| Wet Deposition Flag        | Resistive Raindrop Sensor Boards A/B | Analog ADC / GPIO34 (3.3V Logic)          | Dual raindrop sensor configuration for comparative stability, detecting surface wet deposition and active wash-out events.                                |
| Field Validation Tool      | UNI-T UT363 Digital Anemometer       | Physical Field Gauge (0.1 m/s accuracy)   | Provides on-site wind vector cross-validation against ERA5 meteorological reanalysis during monthly operational maintenance visits.                       |

### 3. Electrical Interface and Pin Wiring Reference

| Device / Subsystem | Module Pin                                                 | Board Pin                                                 | Voltage / Bus                 | Functional Description & Wiring Instructions                                                                                                 |
|--------------------|------------------------------------------------------------|-----------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Plantower PMS7003  | Pin 4 (TXD)<br>Pin 5 (RXD)<br>Pin 1,2 (VCC)<br>Pin 3 (GND) | GPIO16 (RX2)<br>GPIO17 (TX2)<br>VIN (5V Rail)<br>GND Rail | 5.0V Power<br>3.3V UART Logic | Hardware Serial UART2 channel. Laser diode and fan draw from 5V rail; UART data lines operate natively at 3.3V logic without level shifting. |
| Bosch BME280       | SDA<br>SCL<br>VCC<br>GND                                   | GPIO21 (SDA)<br>GPIO22 (SCL)<br>3V3 Rail<br>GND Rail      | 3.3V Power<br>I2C Bus (0x76)  | Standard I2C communications. Requires external pull-up resistors (4.7k ohm) if not integrated on the breakout board.                         |

|                    |                                       |                                                 |                         |                                                                                                                           |
|--------------------|---------------------------------------|-------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------|
| MicroSD SPI Logger | MOSI<br>MISO<br>SCK<br>CS / VCC / GND | GPIO23<br>GPIO19<br>GPIO18<br>GPIO5 / 3V3 / GND | 3.3V Power<br>SPI Bus   | Hardware SPI bus configuration with dedicated Chip Select on GPIO5. Supports FAT32 file system for direct CSV extraction. |
| Raindrop Board     | AO (Analog Out)<br>VCC / GND          | GPIO34 (ADC1_CH6)<br>3V3 Rail / GND             | 3.3V Power<br>ADC Input | Analog precipitation reading. Uses ADC1 channel to avoid Wi-Fi radio conflicts associated with ADC2 channels.             |

### 4. Pakistan-Local Vendor Pricing & Bill of Materials (August 2026)

All components are sourced exclusively through authenticated Pakistan specialist vendors to eliminate international shipping delays, import tariffs, and currency risks. Sourcing is formally cross-verified across **Embedded Studio**, **Digilog.pk**, **Electrobes**, **A.E Solution**, **Clopal Online**, **Expert Tools World**, and **Multan Electronics**.

| #  | Component Specification                                          | Sourcing Vendor             | Qty       | Unit (PKR) | Total (PKR)       |
|----|------------------------------------------------------------------|-----------------------------|-----------|------------|-------------------|
| 1  | PMS7003 PM1/PM2.5/PM10 Laser Sensor                              | Embedded Studio             | 1         | 4,200      | 4,200             |
| 2  | ESP32 WROOM-32D Development Board                                | Digilog.pk                  | 1         | 1,160      | 1,160             |
| 3  | BME280 Temp / Humidity / Pressure                                | Electrobes                  | 1         | 850        | 850               |
| 4  | Arduino MicroSD Card Reader Module                               | Electrobes                  | 1         | 150        | 150               |
| 5  | Samsung EVO Plus 16GB MicroSD Media                              | Daraz.pk Official           | 1         | 1,450      | 1,450             |
| 6  | Waterproof IP65 Electrical Enclosure                             | A.E Solution                | 1         | 750        | 750               |
| 7  | 20cm Mixed Jumper Wire Kit (120 pcs)                             | Daraz.pk                    | 1         | 440        | 440               |
| 8  | 5V 2A AC/DC Power Adapter Unit                                   | Electronics Hub             | 1         | 170        | 170               |
| 9  | Black Nylon Self-Locking Cable Ties                              | Daraz.pk                    | 1         | 120        | 120               |
| 10 | 2-inch Stainless-Steel Hose Clamps (2 pcs)                       | DreamsMart.pk               | 1         | 60         | 60                |
| 11 | Solar Radiation Shield / DIY Stevenson                           | Local source required       | 0         | TBD        | TBD               |
| 12 | Raindrop Detection Module (Rain Sensor A)                        | Digilog.pk                  | 1         | 160        | 160               |
| 13 | Rain Drop Moisture Module (Rain Sensor B)                        | Electrobes                  | 1         | 150        | 150               |
|    | <b>CORE BUILD PRICED SUBTOTAL (Items 1 to 10, 12 to 13)</b>      | <b>12 Items</b>             | <b>12</b> |            | <b>PKR 9,660</b>  |
| 14 | UNI-T UT363 Digital Anemometer                                   | Electrobes                  | 1         | 4,250      | 4,250             |
| 15 | Clopal 10m Heavy-Duty 5-Way Extension Lead                       | Clopal Online               | 1         | 3,295      | 3,295             |
| 16 | GMSA RTV Silicone Caulk Sealant (310ml)                          | Expert Tools World          | 1         | 390        | 390               |
| 17 | M3-M6 Mild-Steel Nuts & Bolts (8 pcs)                            | Multan Electronics          | 8         | 24         | 192               |
|    | <b>VALIDATION &amp; ADDITIONS SUBTOTAL (Items 14 to 17)</b>      | <b>4 Items</b>              | <b>4</b>  |            | <b>PKR 8,127</b>  |
|    | <b>BASELINE HARDWARE SUBTOTAL (Items 1 to 10, 12 to 17)</b>      | <b>16 Items Complete</b>    | <b>16</b> |            | <b>PKR 17,787</b> |
|    | Contingency Reserve (~5% for minor price movements)              | Budget Reserve              | 1         | 889        | 889               |
|    | <b>TOTAL BASELINE BUDGET REQUEST FOR WORK</b>                    | <b> Islamabad / Karachi</b> | <b>17</b> |            | <b>PKR 18,676</b> |
| 18 | <b>Alternative: Camelion CMS-178 10m Reel (replaces Item 15)</b> | Powerhouse Express          | 1         | 4,200      | <b>PKR 20,000</b> |

## 5. 10-Year Continuous Ground-Truth & Multi-City Training

To establish regulatory-grade forecasting accuracy, AirSense models are trained on a unified **10-Year Continuous Hourly Dataset (2015 to 2025)** comprising **578,592 continuous hours** across Pakistan. The dataset synthesizes ECMWF ERA5 hourly meteorological reanalysis, official US Embassy Met One BAM-1020 regulatory stations (2019 to 2025), and Copernicus CAMS atmospheric modeling.

| City / Monitoring Node  | Total Hours (2015-2025) | Avg PM2.5    | Peak PM2.5  | Primary Role in Machine Learning Pipeline                                                                |
|-------------------------|-------------------------|--------------|-------------|----------------------------------------------------------------------------------------------------------|
| Islamabad Campus        | 96,432 continuous       | 51.78 ug/m3  | 508.0 ug/m3 | Primary deployment hub. Characterized by temperate foothill microclimates and diurnal valley stagnation. |
| Karachi Campus          | 96,432 continuous       | 46.83 ug/m3  | 985.0 ug/m3 | Primary coastal deployment hub. High relative humidity and dynamic marine wind dispersion.               |
| Lahore Station          | 96,432 continuous       | 126.67 ug/m3 | 943.0 ug/m3 | Extreme winter smog training ground. Teaches non-linear models severe temperature inversion dynamics.    |
| Rawalpindi Station      | 96,432 continuous       | 52.20 ug/m3  | 175.3 ug/m3 | Urban traffic corridor reference node. Calibrates vehicle emission surge parameters.                     |
| Faisalabad & Peshawar   | 192,864 continuous      | 80.94 ug/m3  | 461.9 ug/m3 | Industrial and basin topography nodes. Validates nationwide cross-regional model generalization.         |
| Combined Master Dataset | 578,592 continuous      | 73.22 ug/m3  | 985.0 ug/m3 | 11 full calendar years of complete hourly atmospheric forcing.                                           |

## 6. Machine Learning Pipeline & Out-of-Sample Benchmarks

| City      | Model Family      | Train R² | Test RMSE      | Test MAE     | Test MAPE  | Operational Status & Role |
|-----------|-------------------|----------|----------------|--------------|------------|---------------------------|
| Islamabad | Gradient Boosting | 0.7973   | 20.37 ug/m3    | 12.48 ug/m3  | 34.63%     | Production Champion       |
| Islamabad | XGBoost           | 0.7964   | 20.42 ug/m3    | 12.49 ug/m3  | 34.70%     | High-Speed Challenger     |
| Islamabad | Random Forest     | 0.7886   | 20.80 ug/m3    | 12.70 ug/m3  | 34.47%     | Bagged Robust Ensemble    |
| Islamabad | Regression        | 0.7308   | 23.47 ug/m3    | 14.85 ug/m3  | 42.09%     | Linear Baseline Anchor    |
| Islamabad | Isolation Forest  | N/A      | 1,834 Outliers | Contam: 0.05 | 7.6% Flags | Hardware Screener         |
| Karachi   | XGBoost           | 0.7729   | 14.89 ug/m3    | 7.89 ug/m3   | 22.96%     | Production Champion       |
| Karachi   | Gradient Boosting | 0.7667   | 15.10 ug/m3    | 8.04 ug/m3   | 23.70%     | High-Precision Challenger |
| Karachi   | Random Forest     | 0.7566   | 15.42 ug/m3    | 8.02 ug/m3   | 23.13%     | Bagged Robust Ensemble    |
| Karachi   | Regression        | 0.7155   | 16.67 ug/m3    | 9.69 ug/m3   | 30.87%     | Linear Baseline Anchor    |
| Karachi   | Isolation Forest  | N/A      | 913 Outliers   | Contam: 0.05 | 3.8% Flags | Hardware Screener         |
| Lahore    | Gradient Boosting | 0.8338   | 42.40 ug/m3    | 24.15 ug/m3  | 41.85%     | Extreme Inversion Model   |
| Lahore    | XGBoost           | 0.8332   | 42.48 ug/m3    | 24.00 ug/m3  | 41.23%     | High-Speed Challenger     |

## 7. Operational Decision Layer & Campus Action Tiers

| PM2.5 Forecast Range    | Air Quality Category           | Automated Campus WQAC Action                                                 | Student & Athletic Advisory Protocol                                      |
|-------------------------|--------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Below 35 ug/m3          | Good / Acceptable              | 100% fresh air dampers open. Normal filtration.                              | Unrestricted outdoor sports and campus activities.                        |
| 35 to 75 ug/m3          | Moderate                       | Modulate fresh air dampers to 70%. Activate secondary stage filters.         | Issue advisory for students with diagnosed respiratory conditions.        |
| 75 to 150 ug/m3         | Unhealthy for Sensitive Groups | Recirculation mode enabled (80% recirc / 20% fresh). HEPA filtration active. | Relocate strenuous outdoor sports indoors. Broadcast campus alert banner. |
| Above 150 ug/m3         | Hazardous / Severe Smog        | 100% recirculation mode with positive pressure ionization active.            | Mandate indoor operations. Distribute N95 protective masks at entrances.  |
| High PM + Stagnant Wind | Atmospheric Inversion Surge    | Pre-cool building envelope 2 hours prior to forecast morning peak.           | Pre-emptive early advisory to campus administration.                      |

## 8. Deployment Checklist and Monthly Maintenance Protocol

| Phase                         | Task & Operational Procedure                                                                                                      | Success / Completion Criteria                                                                   |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Phase 1: Lab Bench Test       | Assemble ESP32, PMS7003, BME280, and SD logger on test breadboard. Flash firmware and verify serial output.                       | Continuous 24-hour error-free data stream verified on UART serial monitor.                      |
| Phase 2: Enclosure Assembly   | Mount components in IP65 enclosure. Fit solar radiation shield over BME280. Apply neutral-cure silicone sealant.                  | Enclosure passed splash test; zero moisture ingress detected.                                   |
| Phase 3: Rooftop Installation | Secure unit to rooftop parapet rail using 2-inch stainless hose clamps. Connect Clopal 10m outdoor extension cable.               | Unit securely mounted vertically; live telemetry packets verified in database within 5 minutes. |
| Monthly Maintenance           | Clean PMS7003 optical inlet with dry compressed air. Perform 3 spot wind checks with handheld anemometer. Inspect silicone seals. | Maintenance log signed off; data completeness verified >98.5% for preceding 30 days.            |

|                                                                                                             |                                                                                                          |                                                                                                             |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Islamabad Campus Authorization:                                                                             | Faculty Supervision Authorization:                                                                       | Karachi Campus Authorization:                                                                               |
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